

Mooring Design Study

Soft moorings for the single buoy, the 4-buoy cluster and the 4×4 platform: criteria, selection, verification and installation checks

Scope

Three articles, one flume, two wave bands

Article	Configuration	Mooring points	Unmoored tilt / heave period
Single buoy	1 spar with a heave plate	radial collar, 4 lines	2.761 / 2.561 s
Cluster	4 buoys hinged to a hub, at 45°	4 spars, 4 lines	2.856 / 2.586 s
4×4 platform	16 buoys at 45°, 2.06 m wide	8 row spars, 4 two-leg bridles	2.924 / 2.606 s

Band	Wave height H	Wave period T	Purpose
Operational	0.04, 0.08, 0.12 m	1.4–3.5 s; 0.05 s step across the resonances	motions: FloatSim validation, criterion 6
Extreme	0.2, 0.35, 0.5 m	a subset, steepness $H/\lambda \leq 0.08$	line and anchor loads, offsets, survival

Flume 3.66 m wide, 2.7 m deep. Regular waves, heading 0. Periods kept $\geq 5\%$ clear of the sloshing cut-ons at 1.25, 1.53 and 2.19 s.

Seven criteria, fixed before any design

#	Criterion	Limit
1	Static tilt	calm $\leq 1^\circ$ (confirmed at rev B); mean tilt under waves $\leq 3^\circ$ at the maximum H
2	Separation	surge, sway and yaw periods $\geq 4 \times 3.5 \text{ s} = 14 \text{ s}$ (amplification $\leq 6.7 \%$)
3	No slack	$T_{\text{min}} \geq 0.15 T_0$ with the drift; applied to the operational band (extreme: down-flume slack allowed)
4	Excursion	mean + dynamic surge $\leq 1.0 \text{ m}$; mean sway $\leq 0.10 \text{ m}$; wall clearance $\geq 0.6 \text{ m}$
5	Clearance	line to spar, pin or hub $\geq 0.10 \text{ m}$; line to the local free surface $\geq 0.10 \text{ m}$
6	Interference	heave RAO moored vs free within $\pm 3 \%$; tilt within $\pm 5 \%$ or 0.5° . Design rule: tilt-period shift $\leq \zeta/3$, heave shift $\leq 1 \%$
7	Capacity	$WLL \geq 3 \times \text{max tension}$; stroke $\geq 1.25 \times \text{max stretch}$; anchor $\geq 3 \times \text{the sum of its legs}$

$\zeta/3$ rule: a tilt-period shift $\delta \leq \zeta/3$ keeps the resonant amplitude $\geq 1/\sqrt{1 + 1/9} = 94.9 \%$ of the unmoored peak.

Every option simulated in FloatSim, chosen by a rule

1 · Decks

Moored FloatSim decks per article: flume BEM databases, submerged cord 0.02 N/m as catenary lines.

2 · Settle

Static equilibrium (scipy root): calm tilt and the at-rest tension of every line.

3 · Modes, pulls

Tilt and heave shift vs the unmoored article; surge, sway, yaw periods; rigid pulls give K.

4 · Wave runs

Regular waves, wave-relative Morison drag, the drift sum applied in-run.

5 · Rule → spec

A written rule selects; the spec's tables and figures come from the same JSON.

Drift sum: conservative by construction

The recorded fixed-body drift bound (splash-zone drag plus the Havelock potential term, no shielding, no credit for body motion) at each spar's waterline, plus FloatSim's own mean force on the moving body.

Validity: LEVEL1 kinematics

Small-angle, valid to 0.1 rad (5.7°). Resonant and extreme cases exceed it: their tilts, loads and offsets are indicative until the LEVEL2 rerun.

How the design evolved

Four revisions and one evaluation

Rev	Commit	Design	Finding
A	a91e1d5	Lines at the pin plane, +0.717 m; +60 % pretension (cluster, platform)	Tilt period −9.2 / −8.8 / −9.0 % at the design T0, 6.5–8.6 × the ζ/3 limit: withdrawn
B	fb6ba5b	Attachment sweep: z −0.50 / −0.15 / −0.15 m, low T0	Within ζ/3; but H = 0.5 m drifts 2.5–3.4 m: excursion was missing from the rule
C	4f8a2f8	Two cord sets on the same anchors: operational + extreme (k ×5 / ×6)	Both jobs met; criterion 2 declared for the extreme set
H	17c8e7b	Evaluation: does the operational set need pretension?	Keep V0: taut, modelled lines (no spec change)
C.1	73a516d	Creep allowance: creep table, daily re-check, pull band	Platform margin lasts ~14 min, buoy yaw ~54 min: re-tension rules

Rev B's operational design is unchanged through C.1: it is the operational set.

Why the pin plane failed: coupling $\approx k \cdot h^2$

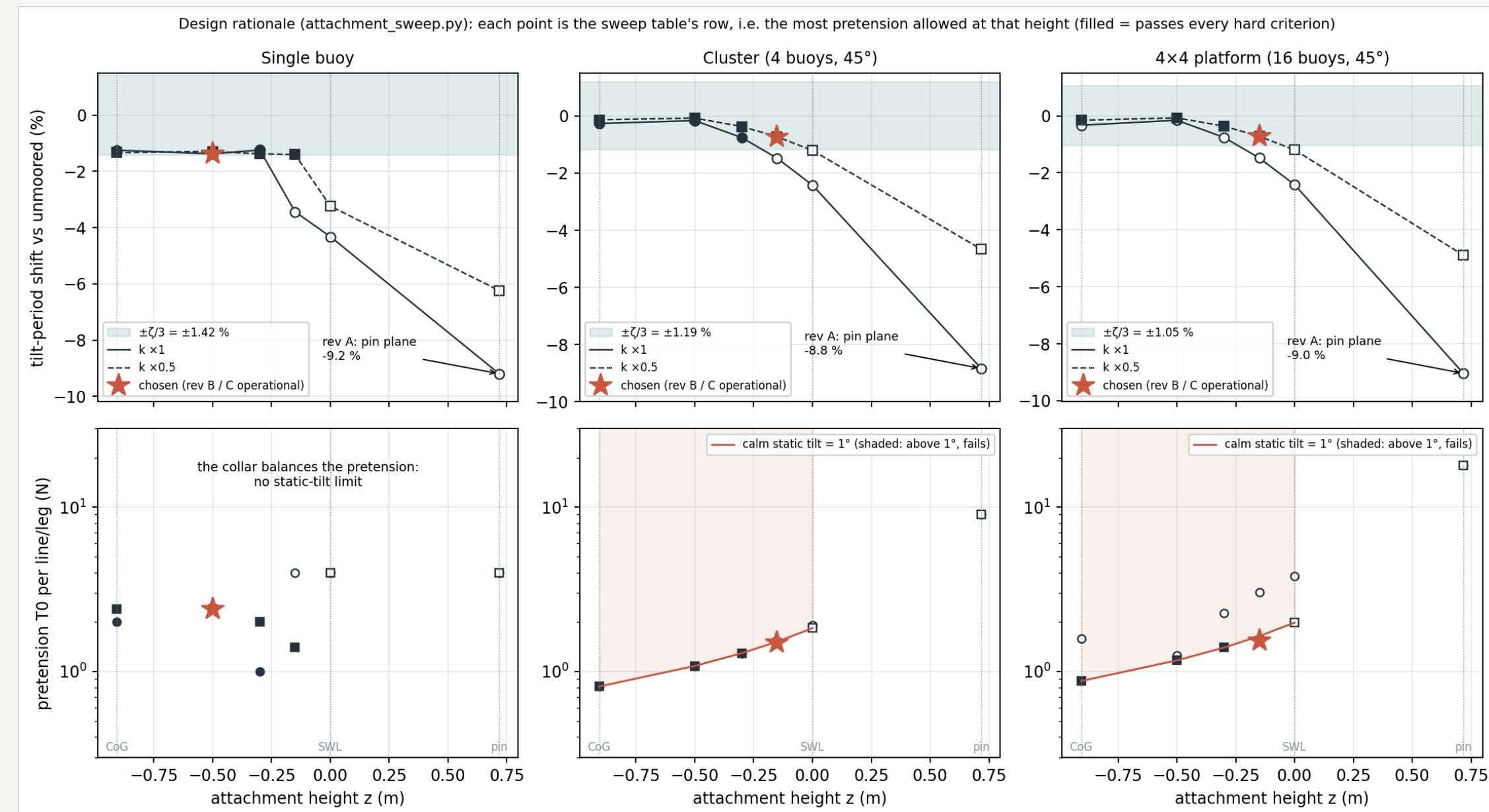
Two effects act about different points:

- The **static** pretension moment acts about the pin (+0.717 m): attaching there gives zero calm tilt.
- The **dynamic** coupling, $\approx k \cdot h^2$, acts about the tilt mode's rotation centre, near the CoG (−0.907 m). At the pin, h is large.
- No single height satisfies both unless T0 is low: attach below the waterline, nearer the rotation centre, with low pretension.

Article	Shift at the pin	Limit $\zeta/3$	Over by
Buoy	−9.19 %	1.42 %	6.5 ×
Cluster	−8.83 %	1.19 %	7.4 ×
Platform	−9.04 %	1.05 %	8.6 ×

ζ : tilt damping at H = 0.04 m, the narrowest band (FloatSim sweep, half-power fit): 4.25 / 3.56 / 3.15 %.

Below the waterline, with low pretension



Hard criteria

- tilt shift $\leq \zeta/3$
- heave shift $\leq 1\%$
- calm tilt $\leq 1^\circ$
- surge ≥ 14 s
- no operational slack, drift sum

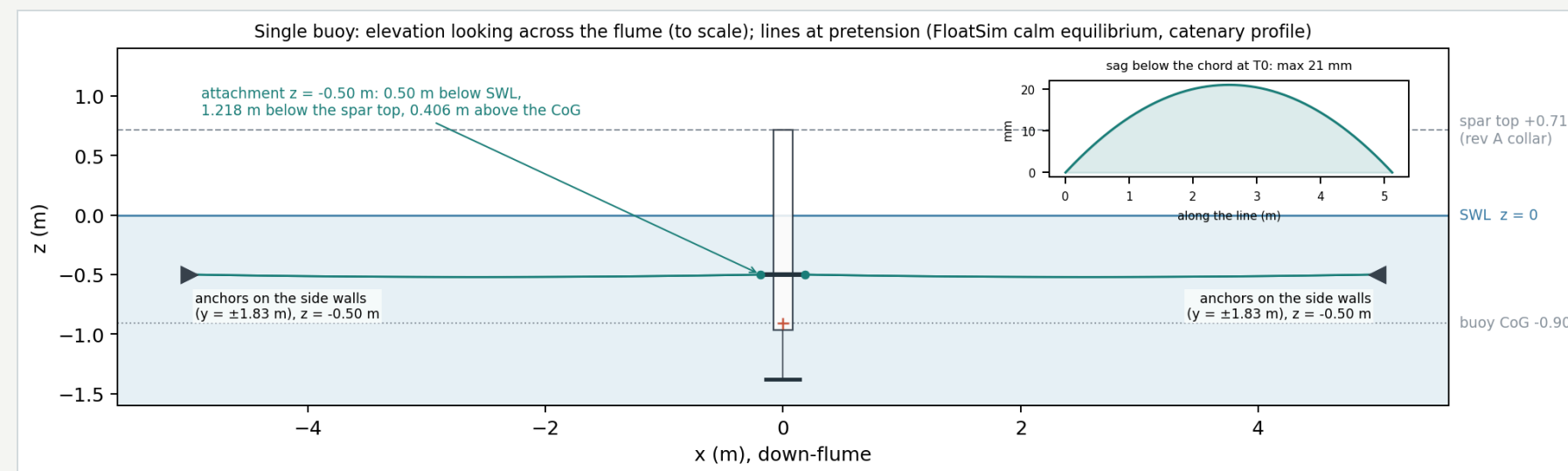
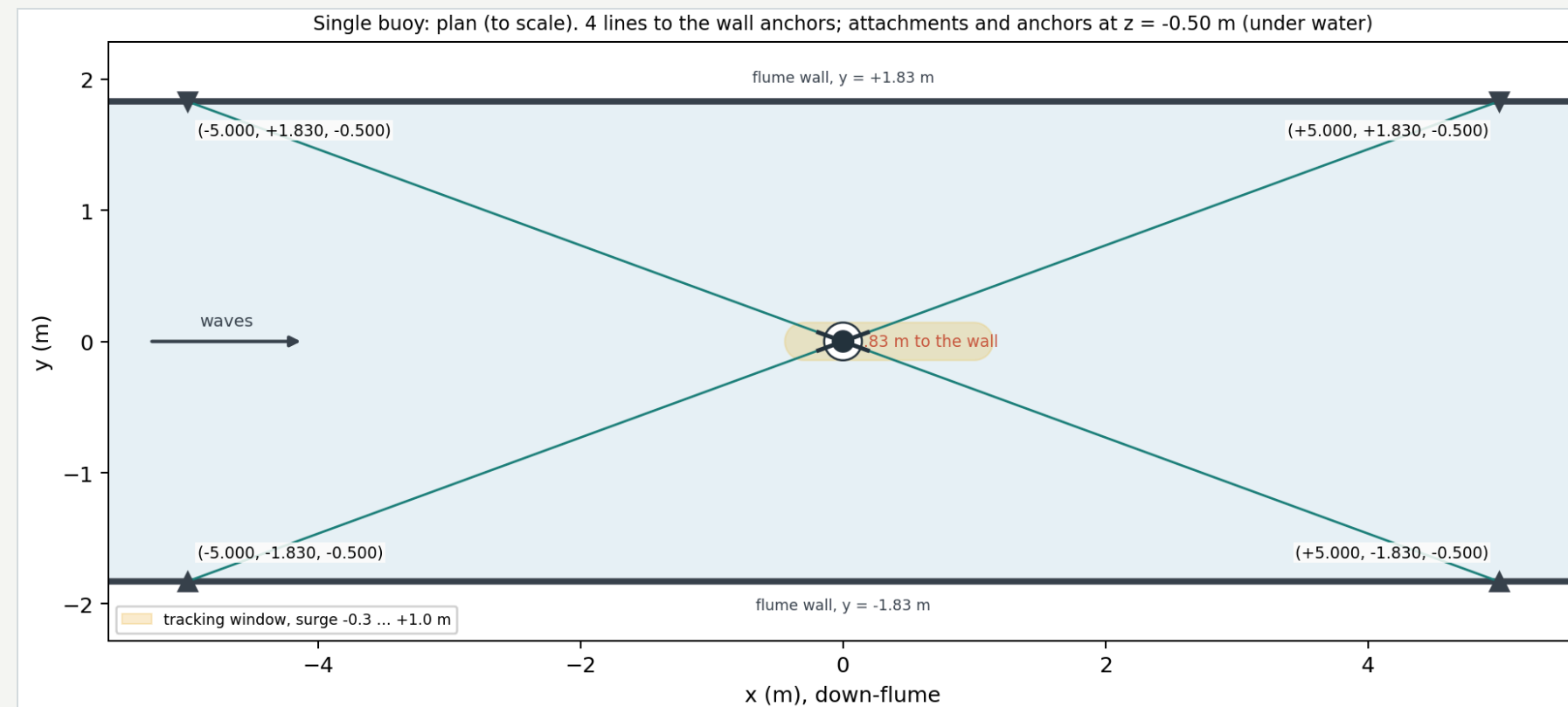
Selection, in order

1. most pretension
2. closest to the waterline
3. stiffest surge (decides the buoy)

Grid: z from the pin to the CoG, $k \times 1$ and $\times 0.5$; each point is the most T0 that passes.

The operational set, one design per article

Quantity	Single buoy	Cluster	4×4 platform
Attachment z	−0.50 m (collar r 0.2 m)	−0.15 m	−0.15 m
Cord k (× design)	2.01 N/m (×1)	4.15 N/m (×0.5)	8.32 N/m (×0.5)
T0 per line, nominal / at rest	2.40 / 2.40 N	1.49 / 1.44 N	1.54 / 1.42 N
L0 · at-rest stretch	3.930 m · 1195 mm	4.589 m · 346 mm	4.038 / 4.208 m · 171 mm
Tilt period, free → moored	2.761 → 2.723 s	2.856 → 2.835 s	2.924 → 2.902 s
Tilt shift (limit ζ/3)	−1.37 % (1.42)	−0.73 % (1.19)	−0.72 % (1.05)
Heave shift	−0.50 %	−0.08 %	−0.03 %
Calm tilt, FloatSim settle	0.00°	1.00°	1.00°
Surge / sway / yaw period	16.6 / 26.4 / 1.12 s	23.1 / 51.6 / 24.8 s	23.0 / 59.0 / 33.4 s
Operational T_min / T0, FloatSim	0.93	0.56	0.19
Surge pull stiffness K	7.31 N/m	15.05 N/m	60.63 N/m



Configuration 1 of 3 • to scale

Single buoy

Radial collar, $r = 0.2$ m, at $z = -0.50$ m. It balances the pretension: calm tilt 0° .

Anchors, side walls

$x \pm 5.0$ m, $y \pm 1.83$ m, $z -0.50$ m

Cord, one set

k 2.01 N/m, 2.40 N at rest, 1195 mm

Tilt period, free → moored

$2.761 \rightarrow 2.723$ s (-1.37% ; $\zeta/3$ 1.42 %)

Surge / sway / yaw

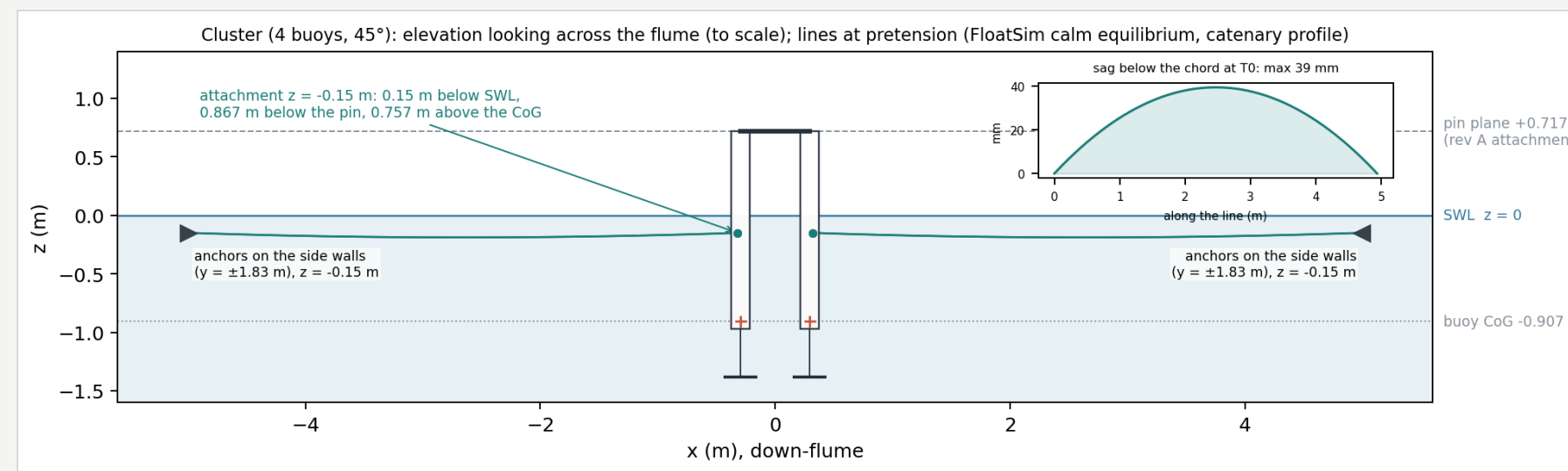
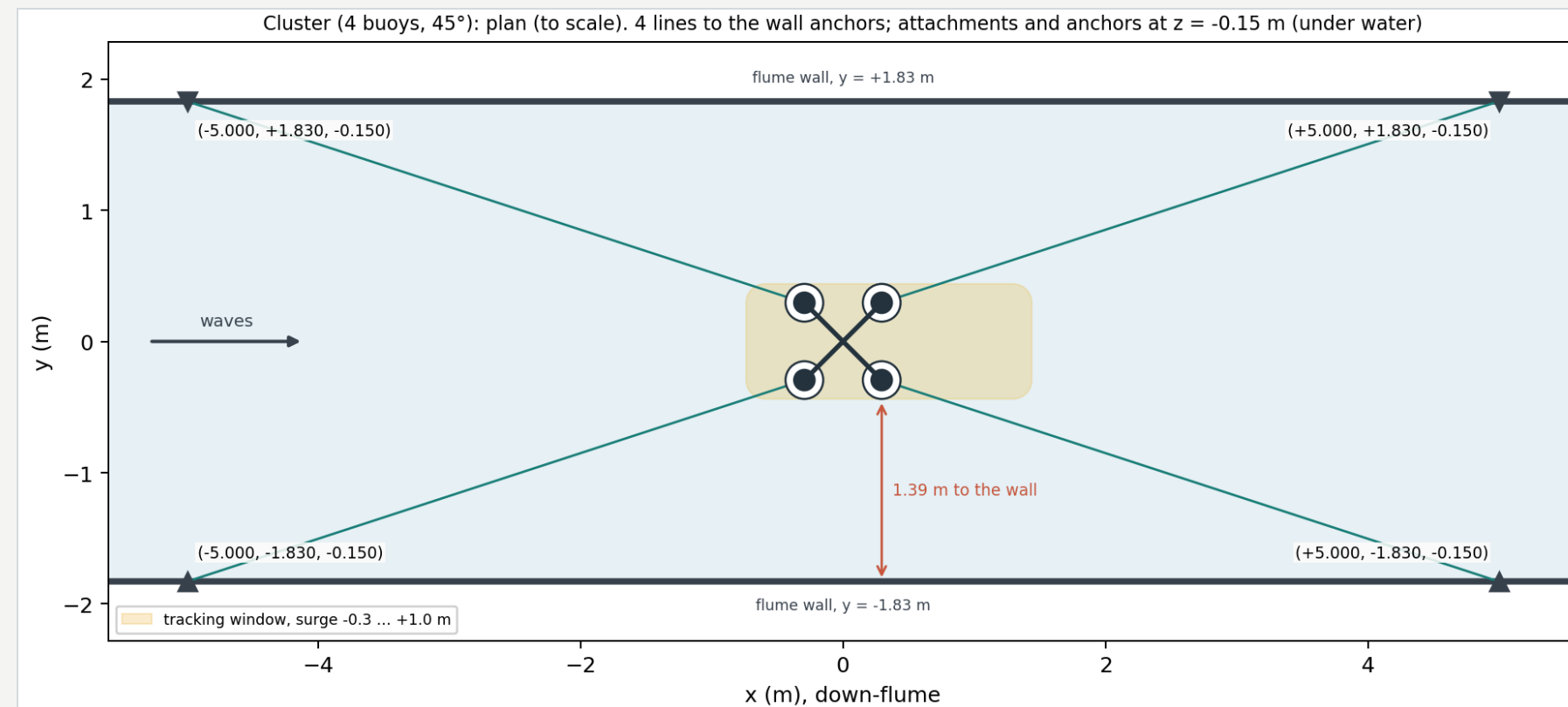
16.6 / 26.4 / 1.12 s

Peak line tension • anchor WLL ×3

3.54 N • 10.6 N

H = 0.5 m max surge

0.68 m, predicted cases: one set



Configuration 2 of 3 · to scale

Cluster

4 buoys hinged to a hub at 45°; one line to each spar at $z = -0.15$ m.

Anchors, side walls

$x \pm 5.0$ m, $y \pm 1.83$ m, $z -0.15$ m

Operational cord

k 4.15 N/m, 1.44 N at rest, 346 mm

Extreme cord, $k \times 5$

$k \geq 20.75$ N/m, 69 mm stretch at rest

Tilt period, free → moored

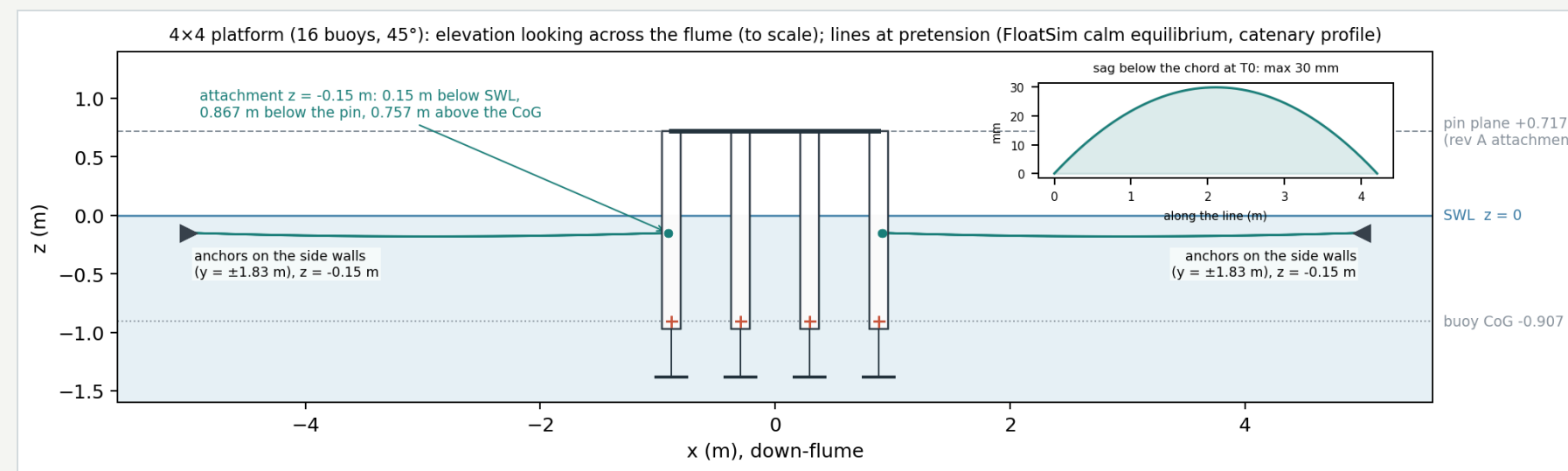
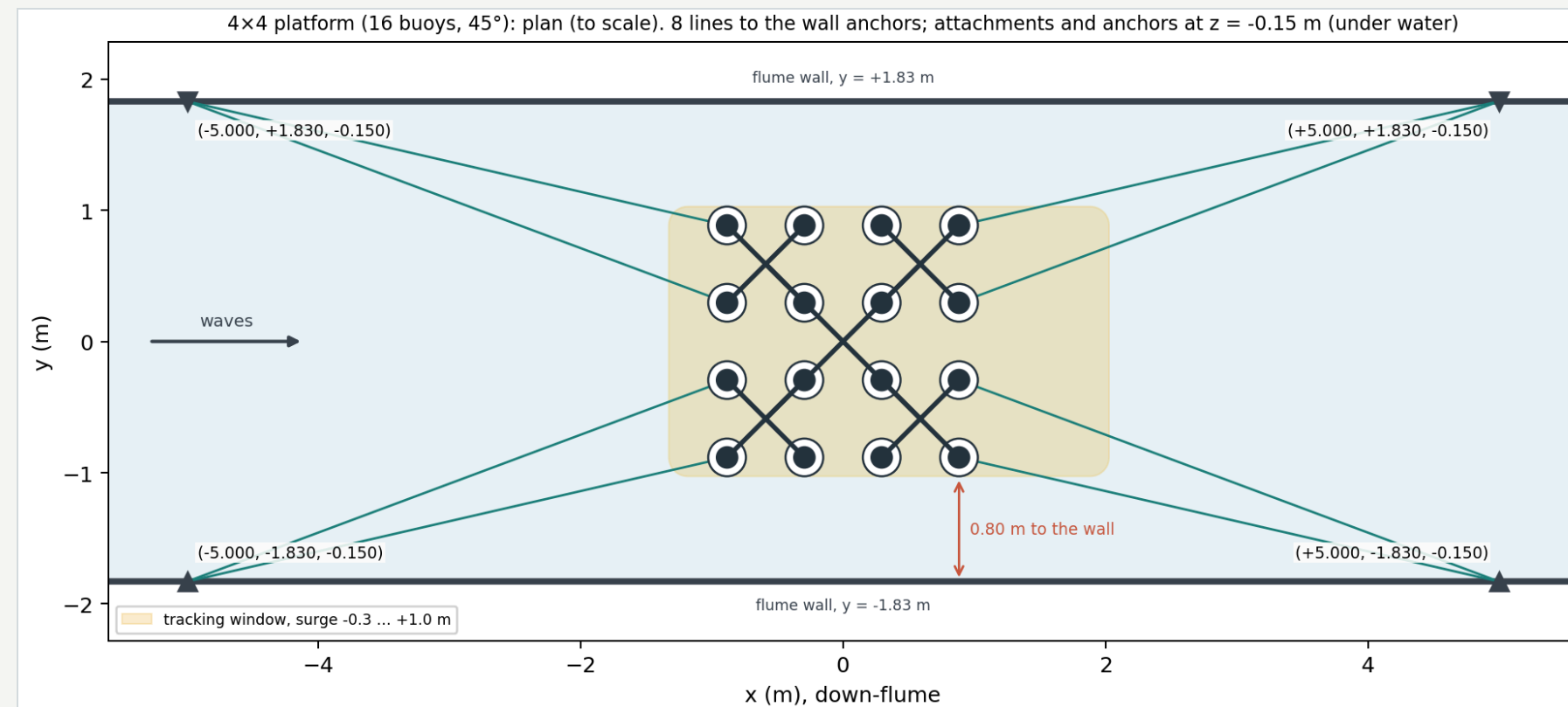
2.856 → 2.835 s (−0.73 %; $\zeta/3$ 1.19 %)

Calm tilt · wall clearance

1.00°, both sets · 1.39 m

H = 0.5 m max surge · anchor WLL $\times 3$

0.97 m, extreme set · 66.4 N



Configuration 3 of 3 · to scale

4x4 platform

16 buoys at 45°, 2.06 m wide. Four 2-leg bridles to the 8 up- and down-stream row spars at $z = -0.15$ m.

Anchors, side walls

$x \pm 5.0$ m, $y \pm 1.83$ m, $z -0.15$ m

Operational cord, per leg

k 8.32 N/m, 1.42 N at rest, 171 mm

Extreme cord, $k \times 6$

$k \geq 49.91$ N/m, 28 mm stretch at rest

Tilt period, free → moored

$2.924 \rightarrow 2.902$ s (-0.72% ; $\zeta/3$ 1.05 %)

Calm tilt · wall clearance

1.00° , both sets · 0.80 m

H = 0.5 m max surge · anchor WLL $\times 3$

0.95 m, extreme set · 267.4 N

Operational band: every line stays taut

Article	Case	Line tension min / max	T_min / T0	Surge: mean (range)	Max tilt	Validity
Cluster	H 0.12 m, T 1.4 s	1.04 / 1.83 N	0.73	0.07 (0.05–0.09) m	2.5°	LEVEL1 valid
Cluster	H 0.12 m, T 2.84 s	0.81 / 2.09 N	0.56	0.00 (±0.04) m	17.6°	indicative
Platform	H 0.12 m, T 1.4 s	0.87 / 1.99 N	0.61	0.06 (0.05–0.07) m	2.2°	LEVEL1 valid
Platform	H 0.12 m, T 2.9 s	0.27 / 2.81 N	0.19	0.00 (–0.05–0.06) m	17.5°	indicative
Buoy	H 0.12 m, T 2.45–2.75 s	– / 2.57 N	0.93–0.95	range ±0.07 m	9.5–16.8°	indicative

Thin margin

The platform at its tilt resonance: T_min / T0 = 0.19 against 0.15. A day's creep takes it below the margin, so it is re-tensioned before these cases.

Drift sign at resonance

FloatSim's own mean force is up-wave there (–0.14 / –0.26 N). Dropping it, the kinematic T_min / T0 is 0.539 / 0.157 (cluster / platform): both still pass.

Moored vs free: heave RAO change at the hub

Wave period	Buoy H 0.04	Buoy H 0.12	Cluster H 0.04	Cluster H 0.12	Platform H 0.04	Platform H 0.12
T 1.4 s (RAO ≤ 0.03)	+0.4 %	−3.5 %	−13.5 %	−15.2 %	−30.4 %	−31.2 %
T 2.2 s	+3.2 %	+9.0 %	+0.8 %	+7.8 %	+0.0 %	+3.2 %
Tilt resonance, 2.72 / 2.84 / 2.9 s	−0.2 %	+2.5 %	+3.2 %	+1.9 %	+4.6 %	+2.8 %
T 3.5 s	−2.1 %	−1.7 %	−0.2 %	+0.1 %	−0.0 %	+0.3 %

1.4 s

The heave RAO is 0.005–0.03: the relative changes amount to ≤ 0.2 mm of motion.

2.2 s

Inside the sloshing exclusion (2.08–2.30 s): a simulation case, not a test period. +9.0 / +7.8 % exceed ±3 %.

Tilt

Moored – free peak: +0.97 to +1.08° off resonance, the 1° calm tilt; +0.45° at the cluster and platform resonance.

Status

Indicative: free runs unrestrained, no applied drift. The formal ±3 % check, with the 60 / 120 s restraint, is Phase D.

Single buoy: stable in operation, yaw-limited in extremes

	T 2.0 s	T 2.35 s	T 2.45 s	T 2.55 s	T 2.65 s	T 2.75 s	T 3.0 s	T 3.5 s
H 0.04 m	—	—	—	—	—	stable	—	—
H 0.12 m	—	—	stable	stable	stable	stable	—	—
H 0.2 m	—	stable	—	—	yaw grows	yaw grows	—	stable
H 0.35 m	yaw grows	yaw grows	—	—	diverges	—	yaw grows	stable
H 0.5 m	—	diverges	—	—	diverges	—	yaw grows	stable

Definitions

stable: yaw at round-off ($\leq 1\text{e-}8$ rad). yaw grows: $< 0.002^\circ$ at 210 s. diverges: no prediction. —: not run. 9 stable, 6 grow, 3 diverge.

Mechanism

The collar's pretension stiffens pitch and yaw together ($\propto T_0 \cdot r$). Every design passing $\zeta/3$ has yaw ≥ 1.12 s, near the $T_p/2$ parametric zone (1.36 s).

Handling

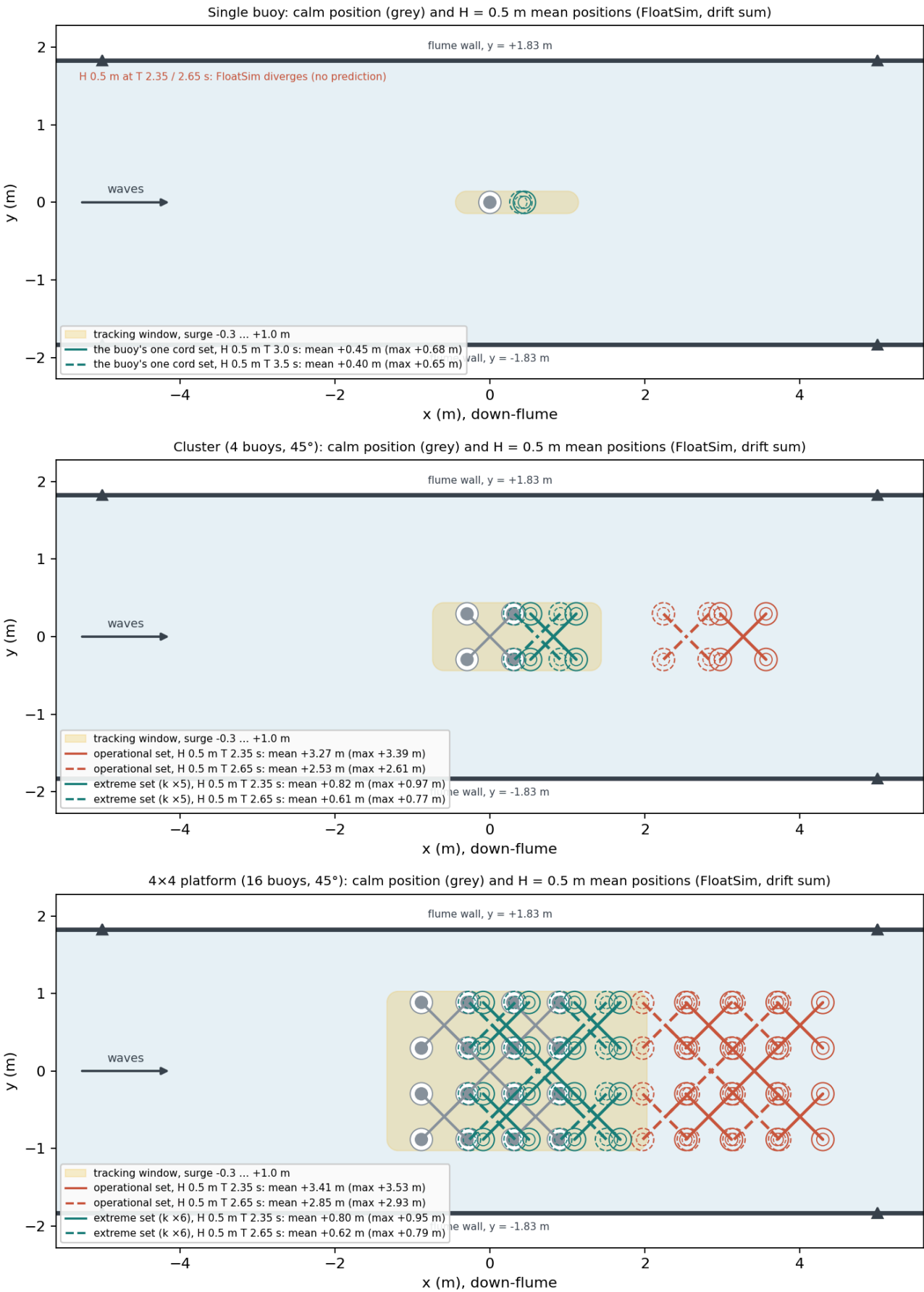
Operational band stable, no slack. Extremes last, with a height ramp; stop on any yaw growth. LEVEL2 decides whether the divergence is physical.

Extreme set • why a second cord set

Rev B's soft set cannot hold H = 0.5 m

Operational set	T	Mean offset (range)	Peak tension
Cluster	2.35 s	3.27 m (3.15–3.39)	15.23 N
Cluster	2.65 s	2.53 m (2.46–2.61)	12.01 N
Platform	2.35 s	3.41 m (3.30–3.53)	29.43 N
Platform	2.65 s	2.85 m (2.78–2.93)	24.70 N

- The down-flume lines go slack and stay slack through the wave train (≤ 0.05 N, their hanging weight).
- Rev B's rule omitted the confirmed excursion criterion 4.
- The conflict is real: interference needs soft lines, station-keeping stiff ones. Two cord sets on the same anchors resolve it.
- The buoy keeps one set: its predicted H = 0.5 m cases reach 0.68 m.



Extreme set • max surge = mean + dynamic, H = 0.5 m, drift sum

The smallest stiffness that keeps max surge ≤ 1.0 m

k	Cluster, T 2.35 / 2.65 s	Cluster	Platform, T 2.35 / 2.65 s	Platform
×3.5	1.30 / 1.02 m	fails	1.41 / 1.15 m	fails
×4	1.17 / 0.92 m	fails	1.29 / 1.05 m	fails
×4.5	1.07 / 0.84 m	fails (mean passes)	1.18 / 0.97 m	fails
×5	0.98 / 0.78 m	CHOSEN	1.09 / 0.90 m	fails (mean passes)
×5.5	0.91 / 0.73 m	pass	1.02 / 0.85 m	fails (mean passes)
×6	0.85 / 0.68 m	pass	0.96 / 0.81 m	CHOSEN
Final	0.97 / 0.77 m (×5)	PASS: the spec	0.95 / 0.79 m (×6)	PASS: the spec

- More than the ~×3 of a 1/k estimate (a rigid-pull estimate gave ×3.5–4): the pretension stiffness does not scale with k, the down-flume lines go slack, and the tilt carries the attachments down-flume.
- Criterion 4 is mean + dynamic; the mean alone would allow ×4.5 / ×5. The band is –0 / +30 %: a minimum stiffness.

Declared effects, and the same tension at rest

Declared, not criteria

Extreme set	Cluster ×5	Platform ×6
Tilt shift vs unmoored	−3.77 %	−4.65 %
Heave shift	−0.09 %	−0.03 %
Surge period	10.5 s	9.6 s
Surge amplification, 3.5 s	+13 %	+15 %

Criterion 2 (≥ 14 s) would cap k at $\sim \times 2.8$ and leave ≥ 1.5 m of offset: 2 and 4 cannot both hold. Criterion 4 governs; accepted as declared, 2026-09-25.

At-rest-matched pretension

The 1° calm tilt moves each attachment $\delta \approx 15$ mm toward its anchor:
 $T_{\text{rest}} = T_0 - k \cdot \delta$, and a stiffer cord loses more.

$$T_{0,\text{ext}} = T_{\text{rest}} + m \cdot (T_0 - T_{\text{rest}})$$

Static residual, operational settle	Cluster	Platform
operational lines	0.0077 N	0.0024 N
extreme, matched T_0	0.0073 N	0.0022 N
extreme, same nominal T_0	0.13 N	0.24 N

Pre-stretch only 69 / 28 mm: set by tension or calm tilt, not by length.

Installation • required before every extreme series

A static pull identifies the installed set

Article	Cord set	Surge pull K	Force at 0.25 / 0.5 m	Extreme if K ≥
Cluster	operational	15.0 N/m	3.7 / 6.4 N	33 N/m
Cluster	extreme ×5	73.2 N/m	12.0 / 21.6 N	33 N/m
Platform	operational	60.6 N/m	12.7 / 20.7 N	146 N/m
Platform	extreme ×6	349.3 N/m	51.0 / 97.6 N	146 N/m

Why a pull

Both sets share the at-rest tension and the 1° calm tilt, so the tilt cannot tell them apart. K differs ×4.9 (cluster) and ×5.8 (platform).

Threshold

The geometric mean of the two sets' K. A day's creep moves each K by ~13 %, far less than the 5–6 × gap.

If it goes wrong

H = 0.5 m on the operational cords: 3.39 / 3.53 m of drift; peak 15.2 / 29.4 N against WLL 6.3 / 8.4 N; stretch 3.67 / 3.54 m against 0.63 / 0.42 m.

Line and anchor loads: the platform's extreme set governs

Article	Cord set	Max line tension	Line WLL ×3	Elongation capacity ×1.25	Anchor load ×3
Buoy	one set	3.54 N	10.6 N	2.21 m (56 % of L0)	10.6 N
Cluster	operational	2.09 N	6.3 N	0.63 m (14 %)	6.3 N
Cluster	extreme ×5	22.12 N	66.4 N	1.33 m (27 %)	66.4 N
Platform	operational	2.81 N	8.4 N	0.42 m (10 %)	16.7 N (2 legs)
Platform	extreme ×6	45.27 N	135.8 N	1.13 m (27 %)	267.4 N (2 legs)

267.4 N

largest anchor working load: platform, up-flume, extreme set

≥ 300 N

specified rating for every wall anchor (F2); the build asserts no load exceeds it

×3 · ×1.25

load and stroke factors: unmeasured Cd, drift as a bound, snatch, fatigue

Cord, anchors and the buoy collar

Elastic cord, usable submerged

Secant stiffness held over the working stretch: operational $-13 / +30$ %, extreme $-0 / +30$ % (a minimum). Submerged weight ≈ 0.02 N/m. Hysteresis is not modelled; the pull test measures it.

Buoy collar: a slender spreader

Round section ≤ 25 mm (added-mass budget 0.4 kg over 0.8 m of arms). A 10 mm rod adds 0.06 kg at ≈ 7 MPa root bending. A solid disk would add ≈ 21 kg (heave plate ≈ 7.9 kg): rejected.

Wall anchors

Four per article, at the attachment depth, $x = \pm 5.0$ m, $y = \pm 1.83$ m; every anchor ≥ 300 N. Underwater wall fixing is facility assumption F1: to confirm with HWRL.

Rejected: pulley and counterweight

Constant tension makes all the restoring geometric: far too soft in surge.

Pretension is kept for model fidelity, not position

Cluster / platform	V0, rev C	V1, taut minimum	T0 = 0	V2, 10 mm slack
At rest per line or leg	1.44 / 1.42 N	0.36 / 0.45 N	0.21 / 0.22 N	0.20 / 0.20 N
Calm tilt	1.00 / 1.00°	0.25 / 0.33°	0°	0°
Tilt-period shift	−0.73 / −0.72 %	−0.54 / −0.60 %	−0.23 / −0.23 %	−0.20 / −0.18 %
Surge pull K	15.0 / 60.6 N/m	11.1 / 49.4 N/m	5.1 / 20.5 N/m	4.5 / 16.1 N/m
Lines at resonance	taut, min 0.27 N	stretch lost 18–40 %	statics only	statics only

Why V0

Taut cords in every operational case, so the tank and FloatSim see the same modelled mooring. V1's slack/taut cycling at resonance is unmodelled dynamics in exactly the validation cases. The buoy keeps its pretension anyway: at V1 its yaw is 2.48 s.

What V1 would buy, and cost

Calm tilt 1.0 → 0.3° and 0.1–0.2 % less tilt shift; heave RAO within 1.2 %. Costs: cord cycling, fouling risk, drift ×1.44–1.49, and the cluster's extreme set leaves the window (1.054 m).

Rev C.1 · creep assumed ~4 % of the stretch per decade (wet natural rubber)

Cord creep: a daily re-check against a creep table

$$r(t) = 1 - 0.04 \cdot \log_{10}(t / 1 \text{ min})$$

Since tensioning	Tension left	Calm tilt
1 min	100.0 %	1.00°
10 min	96.0 %	0.96°
1 h	92.9 %	0.93°
8 h	89.3 %	0.89°
24 h	87.4 %	0.87°

Gent, Engineering with Rubber, ch. 7; PMC6728486.
To confirm with the cord's datasheet.

Article, cord set	Re-tension below	Reached after
Cluster, operational	0.52° / 0.74 N	never
Cluster, extreme	0.72° / 1.03 N	never
Platform, operational	0.95° / 1.35 N	~14 min
Platform, extreme	none	—
Buoy, one set	2.23 N	~54 min

Basis: the no-slack margin (operational), the tracking window (extreme), the buoy's yaw 15 % clear of the T_p/2 zone (1.36 s). A reading > 0.05° or 5 % below the table is slip or damage.

Re-tension the platform's operational set before its H 0.12 m cases at 2.7–3.1 s, and the buoy before its extremes.

Installation and test sequence

- 1** Install the operational set; set T0 by the calm tilt ($\approx 0.70^\circ/\text{N}$, cluster and platform) or tension (buoy). Static pull and decays against the FloatSim predictions.
- 2** Response series, $H \leq 0.12$ m, on the operational set.
- 3** Swap to the extreme set: same anchors, attachments and at-rest tension; set it by tension or the 1° calm tilt, never by length.
- 4** Static pull before every extreme series: $K \geq 33$ N/m (cluster) or ≥ 146 N/m (platform) confirms it.
- 5** Extreme series in increasing H, $0.2 \rightarrow 0.35 \rightarrow 0.5$ m; stop near 0.9 m of offset or a line near its working load.
- 6** The buoy's extremes last, with a height ramp; stop on any yaw growth.

Summary

Where each criterion stands at rev C.1

#	Criterion	Operational set	Extreme set
1	Static tilt	calm 1.00° (0° buoy): met; wave mean: Phase D	calm 1.00°: met; wave mean: Phase D
2	Separation	surge 16.6–23.1 s: met; buoy yaw stiff, 1.12 s	surge 10.5 / 9.6 s: declared, accepted
3	No slack	T_min / T0 ≥ 0.19 (FloatSim): met, platform thin	down-flume lines slack in the train: allowed
4	Excursion	mean offset ≤ 0.07 m at H 0.12 m: met	max surge 0.97 / 0.95 m ≤ 1.0 m: met
5	Clearance	lines submerged, 0.09 m under the trough: met	uncovered 0.16–0.19 m in troughs: reported
6	Interference	tilt shift within ζ/3: met; RAO ±3 %: Phase D	tilt –3.77 / –4.65 %: declared
7	Capacity	line WLL 6.3–10.6 N, stroke ×1.25: output	anchors ≥ 300 N (need 267.4 N): specified

met: shown in FloatSim runs of the chosen design. declared: accepted as a stated effect, not a pass/fail criterion. Phase D: the full 442-case matrix, not yet run.

Known limitations of the analysis

Limitation	Consequence and handling
LEVEL1 small-angle kinematics ($\leq 5.7^\circ$)	Resonant and extreme results indicative (186 of 442 cases); LEVEL2 planned
BEM waterline: 12-sided (NT12)	Resonances $\sim 0.04\text{--}0.08$ s long; $\zeta/3$ holds (relative); NT36 ~ 1.7 h BEM
Drift sum: fixed-body bound + own force	Conservative offsets and tensions; could lower the extreme stiffness
Quasi-static line, no line inertia	Snap loads and the return swing not simulated; $\times 3$ margin, load-cell pull
Line exposure in the $H = 0.5$ m troughs	Uncovered by $0.16\text{--}0.19$ m; line drag not modelled
Extreme yaw, cluster and platform	$0.8\text{--}2.6^\circ$, forced by the $1e-5$ kernel asymmetry; growth not established
Regular waves only	No second-order difference-frequency forcing of the slow surge modes
Creep rate assumed, 4 %/decade	Sets the re-tension times; confirm by datasheet or a one-day hold test

Before the test

HWRL confirmations F1–F7	Gate procurement: underwater wall anchors, ≥ 300 N rating, the two cords
Cord creep rate	Datasheet or a one-day hold test in water; sets the re-tension times
BEM regeneration at NT36	~1.7 h for the platform; resonances ~0.04–0.08 s shorter
Phase D, both cord sets	442 cases (147 / 147 / 148); extreme set so far only at H 0.5 m, 2 periods
LEVEL2 rerun, resonant subset	Resonant tilts, loads and offsets; decides the buoy's divergence
Buoy pitch-neutral collar	E.g. a cross-flume bar: yaw stiffness without the pitch term
Snatch loads, drift mechanism	Dynamic line model or a load-cell pull; a moving-body drift model

Where every number comes from

Record	Holds
MOORING-SPEC.md / .pdf, rev C.1	the specification: line tables, figures, checks, procedure
DESIGN-BASIS.md	every decision and its evidence, Phases A–H
attachment_sweep.py → .json	the operational set: the sweep and the rule
extreme_set.py → .json	the extreme set: k scan, final runs, residuals
spec_statics.json, spec_statics_extreme.json	statics: anchors, attachments, T0, pull K, periods
buoy_stability_check.json	the single buoy's 18-case battery
pretension_study.json, PRETENSION-EVAL.md	the pretension evaluation: V0, V1, T0 = 0, V2
viewer_rows_revc/ (48 runs)	moored vs free comparison; the 3D motion viewer
mooring_figures.py → figs/	the plan, elevation, rationale and offsets figures